Course program and reading list

Semester 1 Year 2026

School: Adelson School of Entrepreneurship B.A

Introduction to Data Science

Lecturers:

Mr. Sahar Millis sahar.millis@post.runi.ac.il

Tutors:

Ms. Gabriella Vagner gabriella.vagner@post.runi.ac.il

Teaching Assistants:

Ms. Danielle Gan gan.danielle@post.runi.ac.il

Course No.:	Course Type :	Weekly Hours :	Credit:
2686	Lecture	5	4

Course Requirements :	Group Code :	Language:
Final Paper	262268600	Hebrew

Course Description

This course equips you with practical data science skills to tackle challenges in business, economics, and entrepreneurship.
You'll learn to frame problems, explore and prepare data, build models, evaluate them, and communicate insights that drive decisions. Through hands-on projects, you'll develop real AI applications and presentation skills, and work with industry-standard tools.
Whether you're a data novice or have some experience, you'll graduate with a powerful toolkit applicable to your studies or entrepreneurial pursuits.

Course Goals

The course will cover the following topics.

Unit #	Title	Topics	Assignment
1	Intro to DS & Lifecycle	What is Data Science? Statistics vs. Machine Learning vs. Deep Learning Problem framing and the DS lifecycle (data → model → insight → action) Roles in DS (analyst, engineer, scientist) and industry use cases Responsible data science: ethics, bias, and reproducibility	
2	Data, EDA & Visualization	Data sources and formats (structured, unstructured, APIs, scraping) Data cleaning: missing values, outliers, normalization Descriptive statistics, correlations, spotting trends and anomalies Visual storytelling: effective charts, dashboards, and critiques	EDA Individual Assignment
3	Classification	Introduction to supervised learning and classification tasks Core models: LR, decision trees, k-NN Evaluation metrics: accuracy, precision/recall, F1, ROC/AUC Applications: fraud detection, spam filtering, churn prediction	
4	Regression	Predicting continuous outcomes with regression models Linear and regularized regression (ridge, lasso) Model diagnostics: residual plots, overfitting vs underfitting Error metrics: MSE, RMSE, MAE, R-squared Applications: price prediction, forecasting trends	Classification & Regression Pair Assignment
5	Clustering	Unsupervised learning: grouping without labels Centroid/Connectivity/Density models Algorithms: k-means, hierarchical, DBSCAN Choosing k and evaluating clusters (Elbow, Silhouette) Applications: customer segmentation, product grouping, market research	

6	Machine Learning	Guess Lecture Feature Engineering, Feature Importance Train/test splits, Cross-Validation, data leakage Grid search, Explainability (depends on class progress)	
7	Recommendation Systems	RecSys Algorithms, Cold Start Intro to Embeddings, Similarity Search Engines	Recommendation Group Assignment
8	Deep Learning	Our human brain Neural networks vs. traditional ML Feedforward nets (MLPs), Embeddings Pre-trained models Tabular DL models	
9	Working with Text Data (NLP)	Text Analysis, Visualizations How computers represent language: tokens, vectors, embeddings Classical approaches (TF-IDF, bag-of-words) vs. modern (transformers) Prompting vs fine-tuning: when and how to adapt language models ROUGE, handling hallucinations Pre-Trained SLM	Final Project, NLP/ CV/Audio Piar
10	Working with Visual Data (Image Analysis)	Our eyes vs how computers "see" pixels, convolution, visual operations Image classification, object detection, and augmentation basics Pre-trained vision models	
11	Working with Multimodal Data	Guess Lecture Combining text-image-audio Intro to audio, cloning, deepfakes Synthetic data generation, data augmentation strategies	

12	Course Summary	Recap Ethical reflections: bias, privacy, responsible AI usage Using AI Agents AutoML, AutoDS (depends on class progress) Feedback & Reflection, course retrospective, Q&A on project progress	
13	Presentations	Final project presentations, Oral defense	Presentation, Oral Exam
14	Presentations	Continuation of project demos and oral exams	Presentation, Oral Exam

- Some adjustments may occur.
- Exact dates for guest lectures will be announced during the course.
- Topics in advanced sessions may be adapted based on the class's progress.



Grading

#	Assignment	Mode	Percentage
1	Exploratory Data Analysis	individual	10%
2	Classification & Regression	individual	20%
3	Recommendation	Pair	20%
4	Final Project & Oral Exam (23/01/2025)	Pair (graded individually)	50%



Learning Outcomes

By the end of the course, you'll be able to:

1. Build real AI applications.
2. Use pre-trained models on text, images, etc.
3. Produce clean, well-documented notebooks and reproducible pipelines.
4. Engineer features, tune models, evaluate models, and interpret results/metrics.
5. Apply and evaluate core models: classification, regression, clustering, and more.
6. Communicate findings from a data-science point of view, both visually and verbally, to stakeholders.

7. Execute an end-to-end DS workflow: problem framing → data acquisition → EDA → modeling → evaluation → storytelling.

AI Policy in the Course

Allowing free use of AI, according to the university's policy

The use of Generative Artificial Intelligence (GAI) tools can offer significant educational value. However, it is important to set clear rules and expectations for their use in assignments and exams.
When using GAI tools in assignments or exams, students must report such use within the submission, specifying how the tools were used — for example, the type of AI tool used or the content of the prompts.
The use of GAI tools in violation of these university guidelines, may be considered a disciplinary offense.

Lecturer Office Hours

Before and after class.
Additionally, upon request - online during the week.

Tutor Office Hours

Online during the week (upon request).

Additional Notes

Course managed through Moodle.

Lesson recordings will be uploaded.

Attendance is aligned with the Entrepreneurship School's guidelines.

About the lecturer

About the Instructor

About the T.A



Reading List

Online:

1. Python.org
2. Scikit-Learn.org
3. Pandas.pydata.org
4. Kaggle.com (Datasets, Learn)
4. HuggingFace.com (Datasets, Models, Spaces)
5. GeeksforGeeks.org (AI, DS, EDA, ML, DL, NLP, CV)

Generative Tools:

1. groq.com
2. claude.ai
3. chatgpt.ai
4. perplexity.ai
5. gemini.google.com

Books:

1. Cole Nussbaumer Knaflic, 2015, "Storytelling with Data".
2. Kieran Healy, 2018, "Data Visualization: A Practical Introduction".
3. Andriy Burkov, 2019, "The Hundred-Page Machine Learning Book".
4. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, 2015, "Deep Learning".
5. Tiffany Timbers, Trevor Campbell, and Melissa Lee, 2024, "A First Introduction".
6. Marek Gagolewski, 2022, "Minimalist Data Wrangling with Python".
7. JEFFREY S. SALTZ JEFFREY M. STANTON, 2024, "An Introduction to DATA SCIENCE".
8. SHAUN AULT, SOOHYUN LIAO, LARRY MUSOLINO, 2025, "Principles of Data Science".

Online Courses:

1. IBM Technology: AI Fundamentals
2. HarvardX: Data Science: Visualization
3. HarvardX: Data Science: Machine Learning
4. HarvardX: Machine Learning and AI with Python
5. HarvardX: Introduction to Data Science with Python
6. HarvardX: CS50's Introduction to Artificial Intelligence with Python
7. MIT, Fall 2016: Introduction to Computational Thinking and Data Science